

TRADINGAGENTS FOREX

TARGET-STATE SYSTEM REQUIREMENTS SPECIFICATION

Practical, implementation-feasible requirements for the finished Forex decision-support platform

AUTHORITATIVE PURPOSE OF THIS DOCUMENT

This document defines what the finished system is required to do. It is NOT a description of what the GitHub repository currently implements. The repository is only a reference implementation baseline. When implementation and this specification disagree, this specification represents the intended target behavior until the requirement is deliberately revised.

Field	Value
System	TradingAgents Forex Decision-Support, Risk, Journal, Backtest and Learning Platform
Reference repository	https://github.com/Zubair815/TradingAgents.git
Reference implementation baseline	main at 87e23b7 (2026-10-01), incomplete by definition
Primary operating environment	Windows, Python, FastAPI, SQLite, MetaTrader 5 / XM Demo or compatible MT5 broker
Execution policy	Human manual execution only; MT5 integration is read-only
Specification status	Target state / normative

Normative terms: MUST / MUST NOT = mandatory; SHOULD / SHOULD NOT = recommended unless a documented reason exists; MAY = optional capability.

Document Map

How an AI coding model should use this specification

Treat requirement IDs as acceptance criteria. Before changing code, map the requested work to the relevant IDs, inspect the existing implementation, identify the gap, implement the smallest coherent change, add regression tests, and report which requirement IDs are satisfied. Do not infer that a requirement is already implemented merely because a similarly named class or endpoint exists.

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1. Purpose, Scope and Product Boundaries

TradingAgents Forex is a single-user or small-team research and decision-support platform that combines market data, specialized LLM agents, deterministic risk controls, manual broker execution, read-only MT5 observation, journaling, analytics, backtesting, and evidence-guarded learning. Its purpose is to improve the quality, traceability and discipline of Forex decisions. It is not an autonomous trading bot.

SCOPE-001 [P0] The system **MUST** produce decision support only: LONG, SHORT, or NO_TRADE, plus a structured proposal and deterministic risk/sizing result when enough evidence exists.

SCOPE-002 [P0] The system **MUST NOT** place, modify, cancel, partially close, or close broker orders. All broker execution remains a human action performed directly in MT5 or another broker interface.

SCOPE-003 [P0] MT5 integration **MUST** remain read-only and limited to connection status, account data, symbol specifications, quotes, candles, positions, pending orders, deals, and observed manual changes.

SCOPE-004 [P0] Deterministic calculations **MUST** have precedence over LLM prose. An LLM **MUST NOT** override a risk rejection, increase an approved lot size, invent a broker value, or fabricate missing market data.

SCOPE-005 [P1] The finished system **SHOULD** support both a browser dashboard and CLI, with the browser being the primary complete workflow for MT5-connected use.

SCOPE-006 [P1] The system **MUST** keep generic stock/crypto capabilities isolated from Forex-specific logic so Forex safety rules do not accidentally alter unrelated asset workflows.

SCOPE-007 [P1] The system **MUST** distinguish implementation completeness, broker-demo acceptance, and trading-performance validation as separate concepts.

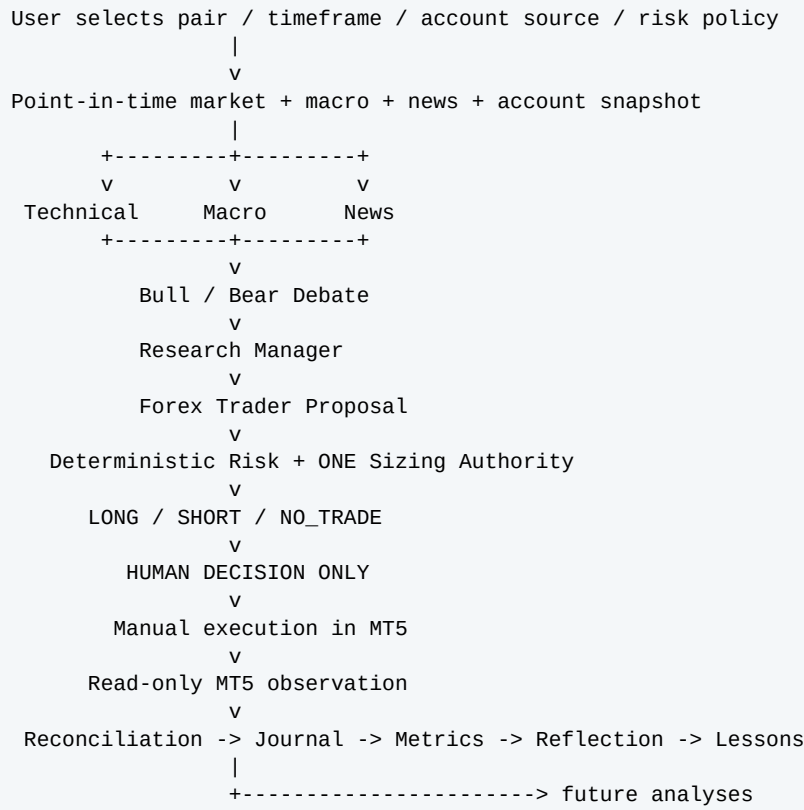
SCOPE-008 [P1] The system **MUST** be usable on a normal Windows machine or Windows VPS without requiring Kubernetes, distributed services, GPUs, or institutional market-data infrastructure.

2. Practicality Filter and Explicit Non-Goals

Requirements in this specification are intentionally constrained to capabilities that are realistic to implement in the existing Python/FastAPI/SQLite/MT5 codebase. The system should be robust, but it should not pretend to be a hedge-fund execution platform.

- PRACT-001 [P0]** The system **MUST NOT** promise profitable, accurate, guaranteed, or statistically proven trading outcomes.
- PRACT-002 [P0]** Historical testing **MUST NOT** be described as tick-perfect unless genuine tick data, broker timestamps, and execution rules are supplied for that test.
- PRACT-003 [P1]** Bar-based historical simulation **MUST** be the default practical model. Lower-timeframe data **MAY** be used to resolve ambiguous intrabar events when available.
- PRACT-004 [P1]** SQLite **MUST** remain an acceptable primary persistence store for the single-user deployment. PostgreSQL or distributed storage **MUST NOT** be required for the initial target state.
- PRACT-005 [P1]** A single Windows process plus local FastAPI dashboard and MT5 terminal **SHOULD** be the preferred deployment. More complex distributed deployment **MAY** be added later but is not required.
- PRACT-006 [P1]** External data that cannot be proven point-in-time **MUST** be marked unavailable for historical analysis rather than reconstructed from present-day values.
- PRACT-007 [P1]** The system **SHOULD** favor a small number of explicit, explainable risk rules over opaque optimization or self-modifying risk policies.
- PRACT-008 [P2]** Multi-hop FX conversion **MAY** be added where needed, but the minimum required implementation is safe direct/inverse conversion for supported account currencies and traded pairs, with fail-closed behavior otherwise.
- PRACT-009 [P2]** Machine-learning models that retrain themselves automatically from live trades are outside the required scope. Learning is evidence retrieval and structured post-trade lessons, not unsupervised strategy mutation.
- PRACT-010 [P2]** High-frequency or sub-second execution is outside scope. The system targets human-reviewed intraday/swing decision support.

3. Target Architecture and End-to-End Flows



ARCH-001 [P0] The system **MUST** assemble one immutable or effectively immutable run context containing the authoritative inputs used by deterministic risk and sizing.

ARCH-002 [P0] The live browser path and CLI MT5 path **SHOULD** reuse the same application-layer risk-context builder rather than duplicating account, quote, broker and portfolio logic.

ARCH-003 [P0] The normal Forex graph **MUST** contain exactly one authoritative final position-sizing calculation for an actionable proposal.

ARCH-004 [P0] A Portfolio Manager or final LLM summary **MUST NOT** override a deterministic NO_TRADE or increase a deterministic approved lot size.

ARCH-005 [P1] The graph **SHOULD** keep independent analysis stages modular so Technical, Macro and News agents can be enabled/disabled without changing risk semantics.

ARCH-006 [P1] All persisted artifacts **SHOULD** carry run_id, snapshot_id or equivalent linkage so the proposal, risk decision, data provenance, execution and lessons can be traced to one analysis run.

ARCH-007 [P1] Transient web presentation state **MAY** be in memory, but proposal/trade/journal/lesson truth **MUST** be persisted outside transient run dictionaries.

4. Core Domain and Shared Contracts

4.1 Pair, action, timeframe and order semantics

DOM-001 [P0] Internally, Forex pairs **MUST** use a canonical six-letter representation such as EURUSD while preserving broker and vendor aliases separately.

DOM-002 [P0] The system **MUST** distinguish LONG, SHORT and NO_TRADE as explicit actions; NO_TRADE is not an error and may be the correct final result.

DOM-003 [P1] Supported order intents **MUST** include MARKET, LIMIT and STOP. Broker execution remains manual even when a proposal contains one of these order types.

DOM-004 [P1] The system **MUST** support first-class execution timeframes and context timeframes. Context timeframes may be higher than the execution timeframe but **MUST** share the same point-in-time cutoff.

DOM-005 [P1] The system **MUST** normalize broker suffixes/prefixes without losing the original broker symbol used for MT5 queries.

4.2 Analysis request

Field	Requirement
pair	Canonical or broker-resolvable pair
execution_timeframe	Primary decision timeframe
context_timeframes	Optional higher/lower contextual timeframes
account_source	mt5 or manual
risk_percent	Requested risk budget before deterministic clamping
research_depth	Controls debate/reasoning depth, not risk rules
selected_analysts	Technical/Macro/News selection
manual_account_*	Required only in manual mode
request_timestamp	UTC request creation time

DOM-006 [P0] account_source **MUST** be explicit and validated as mt5 or manual. A failed MT5 request **MUST NOT** silently become a manual account.

DOM-007 [P0] Manual account mode **MUST** require complete account inputs needed for sizing, including balance, equity, free margin, leverage and account currency.

DOM-008 [P1] Known zero values such as free_margin=0 **MUST** be preserved as zero and **MUST NOT** be replaced by defaults through truthy/falsy shortcuts.

4.3 Canonical run artifacts

DOM-009 [P1] The target contracts **MUST** include AnalysisRun, MarketSnapshot, AgentReport, ForexTraderProposal, ForexRiskDecision, BrokerEvent, TradeRecord, ExecutionRecord, Reflection and Lesson records.

DOM-010 [P1] Records persisted across runs **MUST** include a schema_version and UTC timestamps.

DOM-011 [P1] Model/provider/prompt/strategy version metadata **SHOULD** be stored on proposals or run records so performance can later be segmented by version.

5. Time, Point-in-Time Integrity and Provenance

- TIME-001 [P0]** All internal timestamps MUST be timezone-aware UTC. Naive timestamps MUST be rejected or explicitly normalized at a controlled boundary.
- TIME-002 [P0]** Each analysis MUST have one authoritative cutoff timestamp. No analyst, risk rule, historical query or lesson retrieval may use information known after that cutoff.
- TIME-003 [P0]** Only completed candles may be used as closed-bar evidence. An in-progress candle MUST be clearly identified and MUST NOT be silently treated as complete historical evidence.
- TIME-004 [P0]** Historical news and economic data MUST be filtered using publication/observation/revision timestamps, not merely event dates.
- TIME-005 [P0]** Historical lessons MUST only be available if the lesson itself was created and known before the historical cutoff.
- TIME-006 [P1]** Every externally sourced object SHOULD include source name, retrieval time and observed/published time where applicable.
- TIME-007 [P1]** Conversion observations used for risk MUST include the conversion direction, rate, source and observation timestamp.
- TIME-008 [P1]** Staleness limits MUST be configurable by data type. A stale quote or conversion required for live risk MUST cause a non-actionable result rather than a guessed value.
- TIME-009 [P1]** Session calculations MUST use timezone-aware regional DST rules for London, New York, Tokyo and Sydney instead of fixed UTC offsets.

6. Live Market Data and Quality Controls

- DATA-001 [P0]** MT5 SHOULD be the authoritative live source for broker-facing quote, symbol specification, account state and open portfolio when `account_source=mt5`.
- DATA-002 [P0]** A live quote used for risk MUST include bid, ask, spread and observation time. A mid/last price alone is insufficient for broker-aware spread checks.
- DATA-003 [P0]** Failure to retrieve required live account, position, quote, symbol or conversion data MUST produce `DATA_INSUFFICIENT` or equivalent non-actionable state.
- DATA-004 [P1]** The system MUST support closed OHLCV candles for M1, M5, M15, M30, H1, H4 and D1 where the source can provide them; unsupported resolutions MUST fail explicitly.
- DATA-005 [P1]** Candle series MUST be sorted, deduplicated and validated for invalid OHLC relationships, duplicate timestamps and obviously missing required windows.
- DATA-006 [P1]** Resampled higher-timeframe candles MUST exclude partial buckets at the analysis cutoff.
- DATA-007 [P1]** Yahoo or another fallback price source MAY be used for research/historical context only when explicitly selected or when a configured fallback policy permits it, and the report MUST disclose the source.
- DATA-008 [P1]** The system SHOULD cache external market/macro/news requests with bounded TTLs to reduce rate limits, but cache entries MUST not violate point-in-time semantics.
- DATA-009 [P2]** Volume fields from spot-FX vendors MAY be treated as tick volume or unavailable. The system MUST NOT describe them as centralized exchange volume unless the source actually provides that.

7. Technical Analysis and Market Structure

7.1 Deterministic indicators

- ATR and ATR in pips
- EMA 8/21/34/50/200 and SMA 200
- RSI(14)
- MACD(12,26,9)
- ADX(14) with +DI/-DI
- Stochastic
- Bollinger Bands
- spread/ATR ratio
- distance from 200 EMA
- multi-timeframe alignment

TECH-001 [P0] Indicator arithmetic MUST be deterministic code, not LLM-generated calculations.

TECH-002 [P1] The Technical Analyst MUST receive computed indicator values plus their timeframe and cutoff, then interpret them; it MUST NOT be asked to recreate indicator math from raw text.

TECH-003 [P1] ATR used by deterministic risk MUST come from the deterministic indicator/data layer, not from analyst prose.

TECH-004 [P1] Insufficient warmup bars MUST yield an explicit unavailable indicator rather than a fabricated default.

TECH-005 [P1] Multi-timeframe indicators MUST use only candles completed by the same cutoff.

7.2 Market structure

STRUCT-001 [P1] The system SHOULD support deterministic swing high/low detection and HH/HL/LH/LL structure classification.

STRUCT-002 [P1] Break of Structure and Change of Character detection SHOULD be implemented deterministically from candle structure.

STRUCT-003 [P2] Fair Value Gap and Order Block identification MAY be included as contextual features, but must be presented as deterministic pattern detections rather than proven predictive signals.

STRUCT-004 [P1] Support/resistance proximity and multi-timeframe structure alignment SHOULD be available to the Technical Analyst and proposal report.

8. Macro, News and Economic-Event Intelligence

- MACRO-001 [P1]** The Macro Analyst SHOULD receive currency-specific policy rates, rate differentials, inflation/growth context, broad USD/risk sentiment proxies and relevant central-bank information when reliable data is available.
- MACRO-002 [P0]** Historical macro actuals MUST only be exposed if their publication/observation time proves they were known at the historical cutoff.
- MACRO-003 [P1]** Missing macro data MUST be reported as unavailable; the analyst MAY reason from the available subset but MUST not invent missing releases.
- NEWS-001 [P1]** News records SHOULD include publisher/source, headline, publication time, URL/identifier, currencies, category and relevance metadata.
- NEWS-002 [P1]** Duplicate news items SHOULD be removed by URL and normalized headline similarity.
- NEWS-003 [P0]** High-impact economic events relevant to either currency in the pair MUST be available to deterministic news-blackout risk logic when an event calendar is configured.
- NEWS-004 [P1]** The calendar MUST distinguish scheduled event time from actual publication/revision time where the source supports it.
- NEWS-005 [P1]** If the configured economic-calendar source is unavailable during a policy that requires blackout verification, the system SHOULD fail closed for actionable proposals or clearly apply an explicitly configured degraded mode.
- NEWS-006 [P2]** The system MAY support archive import for historical calendar/news evidence, provided each imported record carries provenance and observation timestamps.

9. Multi-Agent Research Pipeline

Role	Target responsibility
Technical Analyst	Interprets deterministic indicators, volatility, structure and timeframe alignment.
Macro Analyst	Interprets monetary-policy, macro and cross-currency fundamental context.
News/Event Analyst	Interprets current news, scheduled events, surprise/risk themes and market narrative.
Bull Researcher	Builds the strongest evidence-based case for upside / LONG.
Bear Researcher	Builds the strongest evidence-based case for downside / SHORT.
Research Manager	Synthesizes analyst and debate evidence into a directional research conclusion.
Forex Trader	Converts research into a structured proposal with levels, invalidation and rationale.
Deterministic Risk Engine	Validates policy, risk and sizing in code; not an LLM decision authority.
Portfolio Manager	Produces final user-facing synthesis but may not override deterministic safety.

AGENT-001 [P0] Agents MUST receive sourced data and structured context; they MUST NOT be encouraged to fabricate prices, economic releases or broker state.

AGENT-002 [P1] Bull and Bear researchers MUST debate the same snapshot and cite the evidence available in analyst reports rather than independently fetching later data.

AGENT-003 [P1] Debate depth MUST be configurable. A default of one Bull/Bear round is sufficient; deeper rounds MAY be enabled at higher token cost.

AGENT-004 [P1] The Research Manager MUST output a structured directional synthesis and may recommend NO_TRADE when evidence is mixed or insufficient.

AGENT-005 [P0] The Forex Trader MUST output a typed proposal schema. Free-form prose alone is not sufficient to enter deterministic risk evaluation.

AGENT-006 [P0] The Portfolio Manager MUST treat deterministic rejection as final NO_TRADE and MUST NOT invent a different approved size.

AGENT-007 [P1] Every agent report SHOULD record role, model/provider, prompt/strategy version, run ID and cutoff for later audit and performance segmentation.

AGENT-008 [P1] LLM failures SHOULD fail the run or stage explicitly; a missing agent response MUST NOT be replaced with fabricated “successful” analysis.

10. Proposal Contract and Lifecycle

Proposal field	Target meaning
pair	Canonical pair
action	LONG / SHORT / NO_TRADE
order_type	MARKET / LIMIT / STOP
setup_type	Enumerated setup family
timeframe	Execution timeframe
entry_price / zone	Planned entry or bounded zone
stop_loss	Required for actionable directional proposals
take_profit_1 / 2	One or more targets; TP2 optional
suggested_risk_percent	Trader suggestion before policy clamping
suggested_lot_size	Populated only after deterministic sizing
confidence	Model confidence; not a probability unless calibrated
confluence_factors	Structured evidence tags
invalidation_condition	Human-readable invalidation rule
reasoning	Research rationale
decision_time / expiry	UTC lifecycle boundaries
provenance	Run/snapshot/model/strategy references

PROP-001 [P0] An actionable LONG/SHORT proposal MUST contain valid entry and stop-loss geometry before risk evaluation.

PROP-002 [P0] NO_TRADE proposals MUST not require artificial entry/SL/TP values.

PROP-003 [P1] Proposal records MUST be immutable in their original generated payload; later risk/user/broker states should be stored as linked lifecycle evidence rather than silently rewriting the original proposal.

PROP-004 [P1] The system MUST support proposal states at least: PROPOSED, APPROVED or MODIFIED, RISK_REJECTED, WAITING_USER, EXECUTED, SKIPPED, EXPIRED, INVALIDATED, SUPERSEDED and CANCELLED.

PROP-005 [P1] A newer valid proposal for the same pair MAY supersede an older unexecuted proposal, but the older record MUST remain auditable.

PROP-006 [P1] Proposal expiry MUST be determined from explicit validity metadata or deterministic policy, not by silently keeping stale proposals actionable forever.

PROP-007 [P0] Marking a proposal EXECUTED MUST represent user-reported or broker-observed execution evidence. A generated proposal alone MUST never create an actual trade record.

11. Deterministic Risk and Position Sizing

Core rule

Risk and sizing are software calculations, not opinions. There must be one authoritative sizing result for each proposal/context. The LLM may suggest risk and levels, but code decides whether those values are permitted.

11.1 Geometry and policy checks

RISK-001 [P0] Directional proposals MUST have entry and stop on the correct side for LONG/SHORT geometry.

RISK-002 [P0] The system MUST calculate stop distance in price and pips using pair/broker pip size.

RISK-003 [P0] The system MUST calculate risk:reward deterministically and enforce a configurable minimum R:R.

RISK-004 [P0] The system MUST reject an actionable proposal when current spread exceeds the configured absolute maximum.

RISK-005 [P1] The system SHOULD reject an actionable proposal when spread/ATR exceeds a configured ratio, provided ATR is available.

RISK-006 [P1] The system MUST enforce proposal age/expiry and market-open/session rules where configured.

RISK-007 [P1] The system MUST enforce high-impact event blackout policy when the required calendar evidence is available; required-but-unavailable calendar evidence must follow fail-closed policy.

RISK-008 [P0] Requested risk percent MUST be clamped by account policy; the LLM cannot exceed configured maximum risk.

RISK-009 [P0] A risk rejection MUST produce `approved_action=NO_TRADE` and no positive approved lot size.

RISK-010 [P1] A risk modification MAY reduce risk/size or require safer levels, but MUST NOT make a rejected trade more aggressive.

11.2 Pip value and currency conversion

FX-001 [P0] Pip value MUST be calculated using pair pip size, broker contract size when known, lot size and account currency.

FX-002 [P0] If quote currency equals account currency, direct pip-value conversion MUST be used without unnecessary conversion data.

FX-003 [P0] If account currency equals pair base, the traded pair price MAY be used for the mathematically valid quote-to-base conversion, e.g. USDJPY in a USD account.

FX-004 [P0] Cross pairs MUST require a valid conversion observation when account currency differs from both required direct arithmetic currencies. Missing required conversion MUST fail closed.

FX-005 [P1] Conversion lookup SHOULD support direct and inverse observations; optional multi-hop conversion MAY be implemented when needed and must record the path used.

FX-006 [P1] Every conversion used for live or historical risk MUST satisfy a configurable freshness/cutoff rule.

11.3 Position sizing and broker normalization

SIZE-001 [P0] There MUST be one authoritative lot-sizing function/service used by the normal Forex graph.

SIZE-002 [P0] Fixed-risk sizing MUST use account equity, approved risk percent, stop distance and account-currency pip value.

SIZE-003 [P0] The broker volume step MUST be applied conservatively by rounding down, not up, so normalization never increases intended risk.

SIZE-004 [P0] Broker `volume_min`, `volume_max` and `volume_step` MUST be respected when broker constraints are known.

SIZE-005 [P0] If calculated safe volume is below broker minimum, the system SHOULD return `NO_TRADE` / unexecutable rather than increase to broker minimum and exceed risk.

SIZE-006 [P0] An explicit `suggested_lot_size` of 0 or less MUST mean no executable size and MUST NOT fall back to an arbitrary lot such as 0.1.

SIZE-007 [P1] Manual sizing endpoints MAY use clearly documented standard-FX assumptions, but MUST label the result `MANUAL_ESTIMATE` / `broker_verified=false`.

SIZE-008 [P1] Price normalization MUST use broker digits/point when known; hard-coded round(price,5) MUST NOT be the universal Forex rule.

11.4 Margin and free-margin protection

MARGIN-001 [P0] Required margin MUST use actual broker contract size, leverage, entry price and necessary account-currency conversion.

MARGIN-002 [P0] Live MT5 sizing MUST use observed free margin or a validated broker-derived equivalent; it MUST NOT substitute balance for unavailable free margin.

MARGIN-003 [P0] If required margin exceeds available free margin after configured safety buffer, the proposal MUST be unexecutable.

MARGIN-004 [P1] Manual mode MUST disclose that margin is an estimate unless complete broker specifications are supplied.

12. Portfolio-Level Risk and Pending Exposure

- PORT-001 [P0]** All open broker positions **MUST** be included in live portfolio risk before approving new exposure.
- PORT-002 [P0]** Unknown portfolio state **MUST NOT** be interpreted as an empty portfolio. A failed MT5 positions query **MUST** fail closed for actionable live risk.
- PORT-003 [P0]** The system **MUST** enforce a configurable maximum number of simultaneously open positions.
- PORT-004 [P0]** The system **MUST** enforce a configurable maximum total stop-risk amount or percent across existing positions plus the candidate trade.
- PORT-005 [P0]** An existing position without a valid stop or with uncomputable risk **MUST** be treated as unknown/unbounded risk for additional directional exposure unless a documented explicit policy says otherwise.
- PORT-006 [P1]** The system **MUST** support currency concentration checks based on stop-risk exposure by base/quote currency rather than simply comparing raw lot counts to a percent threshold.
- PORT-007 [P1]** The system **SHOULD** reduce or reject additional highly correlated exposure using a deterministic correlation policy or configurable pair-group rules.
- PORT-008 [P0]** Daily realized loss protection **MUST** be supported. Once the configured daily loss ceiling is breached, new risk **MUST** be blocked until the policy reset boundary.
- PORT-009 [P1]** Daily P&L calculation **SHOULD** use broker deals including profit, commission, swap and fees, with the reset-day boundary documented and configurable.
- PORT-010 [P1]** Pending orders **MUST** be represented as reserved future exposure for portfolio risk. The default conservative policy **SHOULD** reserve full planned stop risk.
- PORT-011 [P0]** A pending broker order without a usable stop/risk estimate **SHOULD** block unsafe additional exposure under the conservative default policy.
- PORT-012 [P1]** Open positions and pending exposures **MUST** be aggregated without double-counting the same broker ticket/order.

13. Read-Only MT5 / XM Integration

13.1 Connection and account

MT5-001 [P0] The MT5 adapter MUST expose connection status and sanitized diagnostics without exposing the broker password.

MT5-002 [P0] The account snapshot MUST include balance, equity, margin, free margin, leverage and deposit currency when the terminal provides them.

MT5-003 [P1] Account login shown in UI/logs SHOULD be masked.

MT5-004 [P0] Failure to retrieve account_info required for risk MUST be an explicit unavailable/error state.

13.2 Symbol, tick and candle observation

MT5-005 [P0] Symbol specifications MUST include broker symbol, digits, point, contract size, minimum volume, maximum volume and volume step where available.

MT5-006 [P0] Live ticks MUST expose bid, ask and broker timestamp; spread in pips MUST be derived deterministically using broker/pair pip size.

MT5-007 [P1] The observer MUST support broker symbol suffixes such as EURUSDm without changing the canonical internal pair identity.

MT5-008 [P1] Historical/candle reads MUST keep source and timestamp provenance and exclude unfinished future bars relative to the analysis cutoff.

13.3 Positions, orders and deals

MT5-009 [P0] positions_get returning an empty tuple/list MUST mean verified no open positions; a None/error result MUST mean unavailable, not empty.

MT5-010 [P1] Pending orders MUST be observable with ticket, type, symbol, remaining volume, trigger price, SL/TP and setup time where the terminal provides them.

MT5-011 [P1] Deal history MUST expose ticket/order/position IDs, time, direction/type, entry classification, volume, price, commission, swap, fee, profit and reason when provided by MT5.

MT5-012 [P0] The runtime MUST contain no call path to order_send, broker-side order modification, broker-side close, or cancellation.

MT5-013 [P1] Native MT5 calls that can block SHOULD run outside the main async event loop and SHOULD have bounded orchestration timeouts, while acknowledging native IPC itself may not be force-cancellable.

14. Human Manual Execution and Reconciliation

```

Risk-approved proposal
|
v
UI shows levels, approved size and reasons
|
v
USER chooses EXECUTE / SKIP / WAIT
|
v
If executing: user manually places trade in MT5
|
v
Observer detects broker position/deal
|
v
Matcher links to proposal or marks manual/unplanned
|
v
Journal lifecycle follows observed broker truth

```

EXEC-001 [P0] The system **MUST** require an explicit user decision before treating an approved proposal as intended for execution.

EXEC-002 [P0] The EXECUTE user action **MUST NOT** itself send a broker order. It only records user intent / workflow state.

EXEC-003 [P1] Broker-observed positions **SHOULD** be matched to proposals using exact IDs where available or deterministic evidence such as pair, direction, timing, size and price proximity.

EXEC-004 [P1] Ambiguous matches **MUST** require manual confirmation or remain unplanned; the system **MUST NOT** silently attach the wrong proposal.

EXEC-005 [P1] Trades opened without a matching proposal **MUST** be recordable as MANUAL_UNPLANNED so journal truth remains complete.

EXEC-006 [P1] Repeated observation of the same broker position/deal **MUST** be idempotent and **MUST NOT** create duplicate trades or executions.

EXEC-007 [P1] Manual SL/TP changes, partial closes and final closes **MUST** be detected from broker state/history and appended as lifecycle evidence.

15. Journal, Audit Trail and Persistence

- JOURNAL-001 [P0]** SQLite MUST be the authoritative local store for Forex proposals, risk decisions, observed trades, broker deals/executions, lifecycle events, metrics, reflections and lessons unless a future migration is deliberate.
- JOURNAL-002 [P1]** Database writes MUST use stable IDs and uniqueness constraints or equivalent idempotency keys for broker tickets/deals where possible.
- JOURNAL-003 [P1]** Schema evolution MUST use explicit versioning/migrations. A newer application MUST not silently reinterpret old records with incompatible fields.
- JOURNAL-004 [P1]** The timeline MUST be append-only for audit events; correction should add a new event or update explicit mutable state without deleting historical evidence.
- JOURNAL-005 [P1]** Proposal, trade and execution records MUST retain links to originating run/snapshot/proposal IDs when known.
- JOURNAL-006 [P1]** The system MUST preserve data across dashboard restarts and SHOULD support recovery from a process crash without corrupting the SQLite journal.
- JOURNAL-007 [P1]** User-entered notes/tags MAY be editable, but deterministic broker fields and historical execution evidence SHOULD remain protected from casual mutation.
- JOURNAL-008 [P2]** The system MAY provide export to CSV/JSON/Markdown for user-owned analysis, but export must not become a second authoritative database.

16. Post-Trade Metrics and Execution Analytics

- METRIC-001 [P0]** Realized net P&L MUST be derived from authoritative execution/deal data including commission, swap and fees when available.
- METRIC-002 [P1]** The system MUST calculate pips gained/lost and R-multiple using the original planned risk or documented effective risk basis.
- METRIC-003 [P1]** MFE and MAE SHOULD be calculated over the actual holding window using the best available intraday history. If data is insufficient, the metric MUST be unavailable rather than guessed.
- METRIC-004 [P1]** Execution analytics SHOULD include entry deviation/slippage, spread cost, fill delay when measurable, SL/TP changes, manual exits and partial-close behavior.
- METRIC-005 [P1]** The system MUST distinguish proposal/theoretical performance from actual user execution performance.
- METRIC-006 [P1]** Skipped or expired proposals SHOULD be evaluated counterfactually only when adequate post-proposal data exists, and results MUST be labelled theoretical.
- METRIC-007 [P1]** When both SL and TP are touched in the same bar and lower-resolution evidence cannot establish order, the outcome MUST be AMBIGUOUS or resolved conservatively, not assumed profitable.
- METRIC-008 [P2]** Outcome classification MAY identify patterns such as premature exit, standard win/loss, excessive adverse excursion or poor execution, but the classification MUST be deterministic and diagnostic rather than a trading signal.

17. Reflection, Lessons and Memory

- LEARN-001 [P1]** Closed trades SHOULD trigger a post-close processing pipeline that assembles proposal, risk decision, execution, timeline, MFE/MAE and outcome evidence.
- LEARN-002 [P1]** The Reflection Agent MAY use an LLM to explain what happened, but its inputs MUST come from persisted evidence and deterministic metrics.
- LEARN-003 [P1]** Reflections and lesson extraction MUST be retryable if an external LLM/provider fails.
- LEARN-004 [P1]** Lessons MUST be structured with fields such as pair/setup/session/regime, observation, evidence, actionable rule, confidence/evidence class, source trade IDs and timestamps.
- LEARN-005 [P0]** A single trade MUST NOT become a “proven rule”. Lesson strength MUST reflect evidence count and quality.
- LEARN-006 [P1]** Lessons SHOULD use configurable minimum support before they can materially influence future proposals, while lower-support lessons may remain visible as tentative observations.
- LEARN-007 [P1]** Lesson retrieval SHOULD rank relevance by pair/currency, setup, timeframe, session, regime, recency and evidence strength.
- LEARN-008 [P1]** Retrieved lessons injected into future analysis MUST remain advisory context and MUST NOT bypass deterministic risk rules.
- LEARN-009 [P1]** Historical backtests MUST only retrieve lessons known before the historical cutoff.
- LEARN-010 [P2]** Lesson relevance SHOULD decay with age unless reinforced by newer evidence. Decay parameters should be configurable rather than hard-coded as universal truth.

18. Performance Analytics and Calibration

- AN-001 [P1]** Performance reports SHOULD include trade count, win/loss/breakeven count, win rate, average/median R, expectancy, profit factor, gross/net P&L, drawdown, streaks and holding duration.
- AN-002 [P1]** Performance SHOULD be segmentable by pair, timeframe, setup, session, direction, weekday, regime, news condition, provider/model/prompt/strategy version and confidence band where sample size permits.
- AN-003 [P1]** Metrics MUST include sample-size warnings and MUST avoid strong conclusions from small samples.
- AN-004 [P1]** Raw LLM confidence MUST NOT be displayed as a calibrated probability unless empirical calibration supports that interpretation.
- AN-005 [P1]** Confidence calibration SHOULD bucket proposal confidence and report observed count, win rate, average R and expectancy with explicit small-sample warnings.
- AN-006 [P2]** Monte Carlo bootstrap MAY estimate drawdown/terminal-wealth distributions from historical trade outcomes, but MUST be labelled a stress test dependent on the supplied sample, not a forecast guarantee.
- AN-007 [P2]** Stop/target calibration MAY use historical MFE/MAE evidence to suggest research parameters, but changes to production defaults SHOULD require human review.
- AN-008 [P1]** Ablation studies MUST compare variants on the same dataset/time windows and MUST remain descriptive unless a statistically valid study is explicitly performed.

19. Historical Backtesting

Historical design principle

The backtester is a causal replay of the decision-support workflow, not a price-chart simulator that lets the AI see the future. A decision generated after bar T closes cannot execute at a price that existed before that decision.

- BT-001 [P0]** Historical analysis **MUST** iterate chronologically and expose only data available at each cutoff.
- BT-002 [P0]** A proposal generated from completed bar T **MUST NOT** execute on bar T. A MARKET proposal becomes eligible at the next eligible market price after the decision timestamp.
- BT-003 [P0]** LIMIT/STOP proposals **MUST NOT** trigger before their creation/decision timestamp.
- BT-004 [P0]** If no future eligible price exists, the final proposal remains unfilled rather than being force-filled at the last observed close.
- BT-005 [P0]** The backtester **MUST** never invent a positive lot size when the deterministic approved size is None, zero or negative.
- BT-006 [P1]** Historical simulation **SHOULD** model bid/ask spread, slippage, commission, swap and leverage/margin with explicit assumptions when exact broker history is unavailable.
- BT-007 [P1]** The simulator **MUST** use evolving balance, equity, used/free margin and open positions, not a constant starting account snapshot for every trade.
- BT-008 [P1]** Historical portfolio risk **SHOULD** use the same deterministic risk/sizing rules as live mode with historical point-in-time context.
- BT-009 [P0]** Cross-currency pip value, P&L and margin **MUST** use conversion observations known at the simulated timestamp. Future/current conversions are forbidden.
- BT-010 [P1]** Historical ATR **SHOULD** be calculated from cutoff-safe warmup candles and supplied to spread/ATR risk checks when enough data exists.
- BT-011 [P1]** Historical broker constraints **MAY** be configured as assumptions and **MUST** be labelled as configured/estimated unless sourced from verified historical broker specifications.
- BT-012 [P1]** When same-bar SL and TP ambiguity remains after available lower-timeframe evidence, the simulator **MUST** mark the result ambiguous or use a documented conservative rule.
- BT-013 [P1]** Historical results **MUST** retain `validated_strategy_performance=false` or equivalent until a separate statistical validation process justifies otherwise.
- BT-014 [P1]** The system **MUST** report data provenance, cost assumptions, unresolved ambiguities and unavailable historical evidence alongside backtest results.

20. Walk-Forward, Out-of-Sample and Ablation

- WF-001 [P1]** Walk-forward validation SHOULD use chronological development and out-of-sample windows with no future leakage.
- WF-002 [P1]** An expanding-window design is the preferred practical default: each later development period may include earlier observed history but never its own future OOS segment.
- WF-003 [P1]** Warmup bars before an OOS period MAY be used for indicators, but MUST NOT be counted as OOS trades or learned from as future outcomes.
- WF-004 [P1]** The complete requested OOS tail SHOULD be accounted for; data should not be silently discarded because split lengths divide unevenly.
- WF-005 [P1]** Return-based walk-forward efficiency MUST be unavailable when mean development return is zero or negative; the system MUST NOT invent an epsilon denominator.
- WF-006 [P1]** Labels such as robust/marginal/overfitted, if retained, MUST be described as heuristics rather than statistical proof.
- WF-007 [P1]** Ablation variants MUST share identical input data and period boundaries so differences are attributable to the feature change rather than sample changes.
- WF-008 [P2]** Automated hyperparameter optimization MAY be added later, but if used it MUST be restricted to development data and frozen before OOS evaluation.

21. Web Dashboard and User Experience

View	Required contents
Dashboard	Account/MT5 status, recent proposals, open positions, key performance, alerts/unavailable states
Analysis	Pair/timeframe/account source/risk inputs, run progress, final report
Proposal	Research, levels, deterministic risk checks, approved size, user action
MT5	Connection state, masked account, positions, pending orders, recent deals
Journal	Trades, timeline, costs, modifications, reflection and lessons
Performance	Realized and theoretical metrics, segmentation, calibration warnings
Learning	Lessons, evidence strength, filters and source-trade navigation
Backtest	Historical run, assumptions, walk-forward and ablation results
Settings	Allowlisted runtime settings and secret-presence status, never secret values
Health	Provider/MT5/runtime diagnostics and explicit degraded states

UI-001 [P0] The UI **MUST** clearly distinguish actionable risk-approved proposals from NO_TRADE, rejected, unavailable and manual-estimate states.

UI-002 [P0] The UI **MUST** state that execution is manual and **MUST NOT** contain a control that directly submits a broker order.

UI-003 [P1] Analysis progress **SHOULD** stream server-side stage events so users can see Technical, Macro, News, Debate, Manager, Trader, Risk and finalization progress.

UI-004 [P1] Cancellation **MUST** stop future report/proposal completion where cancellation wins the persistence ordering; the UI must reflect the server's terminal state.

UI-005 [P1] Unavailable data **MUST** render as unavailable with reason, not as zero, blank-success, or fabricated fallback.

UI-006 [P1] Sensitive values such as API keys and MT5 passwords **MUST** never be rendered back into the page after submission.

UI-007 [P1] The dashboard **SHOULD** preserve journal/lesson data across reloads because those values come from persistent stores, not browser local storage.

UI-008 [P1] Browser authentication/session state **SHOULD** use HttpOnly cookies rather than localStorage tokens for protected Forex routes.

22. Backend API and Application Services

- API-001 [P0]** All Forex API routes that expose user/account/journal operations **MUST** be authenticated by default.
- API-002 [P1]** Request validation errors **MUST** return structured safe error codes/messages without raw stack traces or secrets.
- API-003 [P1]** The backend **SHOULD** expose analysis start/status/events/report endpoints, proposal lifecycle endpoints, MT5 read-only endpoints, journal/metrics/learning endpoints, backtest endpoints and settings/health endpoints.
- API-004 [P1]** Standalone proposal sizing/risk endpoints **MAY** exist for diagnostics, but **MUST** clearly label caller-supplied non-broker-verified assumptions.
- API-005 [P0]** Normal MT5-backed analysis **MUST** build risk context on the server from authoritative MT5/market data rather than trusting browser-submitted live account/broker values.
- API-006 [P1]** Long-running analysis/backtest work **SHOULD** execute outside the request thread/event loop with status/events available to the client.
- API-007 [P1]** Application-layer context builders **SHOULD** be framework-independent enough to be reused by web and CLI rather than embedding all business logic directly in FastAPI route functions.

23. CLI

- CLI-001 [P1]** The CLI MUST identify Forex mode separately from generic stock/crypto analysis and collect pair/timeframe/risk/account-source inputs.
- CLI-002 [P0]** If CLI account_source=mt5, it SHOULD use the same authoritative live risk-context assembly as the web path; if this is unavailable, it MUST fail rather than silently using a fake account.
- CLI-003 [P1]** If CLI account_source=manual, it MUST require complete manual account assumptions and label resulting sizing as non-broker-verified.
- CLI-004 [P1]** CLI output SHOULD show final action, proposal levels, deterministic risk reasons, approved size, data-source summary and manual-execution warning.
- CLI-005 [P1]** API keys entered interactively MAY be stored in a local .env only with explicit user intent; the CLI must never print the secret value back in logs/report output.

24. Security and Secrets

SEC-001 [P0] Secrets MUST be sourced from environment variables or protected runtime input and MUST NOT be committed to the repository.

SEC-002 [P0] .env, broker passwords, API keys, session tokens and SQLite user databases MUST be excluded from Git by default.

SEC-003 [P0] MT5 passwords MUST be request-only/connection-only and MUST NOT be persisted in proposal, journal or report records.

SEC-004 [P1] Browser session cookies SHOULD be HttpOnly and SameSite=Strict; Secure SHOULD be enabled when HTTPS is used.

SEC-005 [P1] CORS origins MUST be allowlisted and default to localhost/local deployment origins.

SEC-006 [P1] File paths derived from ticker/pair/user input MUST be sanitized against traversal outside intended directories.

SEC-007 [P1] Provider and broker exceptions exposed to users MUST be sanitized while full technical details may be logged locally without secrets.

SEC-008 [P1] Configuration/status endpoints MUST return secret presence/status only, never secret values.

SEC-009 [P2] If exposed beyond localhost, the recommended deployment SHOULD use a private overlay such as Tailscale or reverse proxy with TLS rather than opening the raw dashboard publicly.

25. Runtime, Concurrency and Cancellation

- RUN-001 [P1]** Each analysis/backtest MUST have a unique run ID and explicit state such as queued/running/finalizing/completed/failed/cancelled.
- RUN-002 [P1]** Independent transient run state MUST not leak configuration or messages between concurrent runs.
- RUN-003 [P1]** Persistence locks MUST be held only around short state/database critical sections, never around LLM or network calls.
- RUN-004 [P1]** Cancellation and final proposal persistence MUST have a deterministic ordering so a cancelled run cannot later publish a successful final proposal.
- RUN-005 [P1]** Repeated cancellation MUST be idempotent.
- RUN-006 [P1]** SSE/event history MUST be bounded in memory. Pruning transient web state MUST NOT delete persistent journal/proposal/lesson evidence.
- RUN-007 [P1]** Stale active runs SHOULD be detected and transitioned to a safe failed/stale state after a configurable retention window.
- RUN-008 [P2]** Checkpoint/resume MAY be supported for long LLM graph runs, but resume MUST preserve the same run signature and data cutoff.

26. Configuration, LLM Providers and Cost Control

- CFG-001 [P1]** The system **MUST** support a typed runtime configuration with validated ranges for debate rounds, token limits, retries, timeframes, risk defaults and polling intervals.
- CFG-002 [P1]** Configuration precedence **MUST** be deterministic and documented, e.g. defaults < persisted settings < environment < explicit per-run overrides.
- CFG-003 [P1]** The system **SHOULD** support multiple LLM providers through a common client factory, including OpenAI-compatible endpoints and local models when configured.
- CFG-004 [P1]** Quick and deep model roles **SHOULD** be configurable separately to control cost and reasoning depth.
- CFG-005 [P1]** Provider-specific reasoning/thinking settings **SHOULD** be passed only to providers/models that support them.
- COST-001 [P1]** Before an expensive historical/agent run, the system **SHOULD** estimate analysis points, expected LLM calls, estimated input/output tokens and approximate model cost using the actual graph shape and debate rounds.
- COST-002 [P1]** The estimator **MUST NOT** count deterministic risk calculations as LLM calls.
- COST-003 [P1]** Actual LLM/tool call and token usage **SHOULD** be collected when provider metadata is available so estimates can be compared with observed usage.
- COST-004 [P1]** Cost estimates **MUST** disclose that retries, tool calls, cache hits and provider pricing can change actual cost.
- CFG-006 [P2]** Runtime settings changes that require restarting the MT5 polling service **SHOULD** be identified explicitly to the user.

27. Backup, Restore and Disaster Recovery

BACKUP-001 [P1] The system **MUST** provide a safe SQLite journal backup function using SQLite backup semantics or equivalent consistency-safe snapshotting rather than raw copying an actively written file.

BACKUP-002 [P1] Backups **SHOULD** include timestamp and application/schema-version metadata and **MUST** not include .env/API/broker credentials.

BACKUP-003 [P1] A created backup **MUST** pass PRAGMA integrity_check or equivalent before being reported successful.

RESTORE-001 [P1] Restore **MUST** validate that the source is a readable compatible TradingAgents journal before replacing the active database.

RESTORE-002 [P1] Restore **SHOULD** create a safety backup of the current database before replacement.

RESTORE-003 [P1] Restore **MUST** be performed while database writers are stopped or coordinated so no concurrent write can corrupt the restored state.

RESTORE-004 [P1] After restore, the application **MUST** reinitialize database services and verify integrity/migrations before normal operation resumes.

BACKUP-004 [P2] A simple CLI backup/restore command is sufficient for the target state; a complex cloud backup service is not required.

28. Deployment and Operations

Recommended practical deployment

Windows PC / Windows VPS

- | - XM MetaTrader 5 terminal
- | - TradingAgents Python environment
- | - FastAPI dashboard (localhost)
- | - SQLite journal under user data directory
- ` - Optional Tailscale/private access

No Kubernetes. No broker execution service. No mandatory cloud database.

DEPLOY-001 [P1] The preferred production-like personal deployment **MUST** support Windows because the MetaTrader5 Python integration depends on the local MT5 terminal environment.

DEPLOY-002 [P1] The dashboard **SHOULD** bind to 127.0.0.1 by default.

DEPLOY-003 [P1] If remote personal access is required, a private network/tunnel **SHOULD** be preferred over exposing the dashboard directly to the public Internet.

DEPLOY-004 [P1] Application startup **SHOULD** tolerate MT5 being offline: the dashboard may start in degraded observer-offline state, but MT5-backed analysis **MUST** fail closed until connection is verified.

DEPLOY-005 [P1] Graceful shutdown **SHOULD** stop polling threads/services and close database/MT5 resources without leaving partially written state.

DEPLOY-006 [P1] Logs **SHOULD** rotate or remain bounded enough for long-running personal deployment.

DEPLOY-007 [P2] Container deployment **MAY** remain available for generic/non-MT5 workflows, but native Windows deployment is the required path for MT5-connected use.

29. Testing and Validation

- TEST-001 [P0]** Every deterministic financial calculation MUST have focused unit tests covering normal, boundary and failure cases.
- TEST-002 [P0]** Risk-context integration tests MUST prove that live account, spread, ATR, broker constraints, conversions, open positions, pending exposure and daily P&L reach the normal risk/sizing path.
- TEST-003 [P0]** Cross-pair tests MUST include cases such as EURJPY with USD account, not only direct USD-quoted pairs.
- TEST-004 [P0]** Tests MUST verify positions_get None/error fails closed while a verified empty tuple/list is treated as no positions.
- TEST-005 [P0]** Historical tests MUST prove no same-bar post-decision execution and no future-data leakage.
- TEST-006 [P1]** Integration tests MUST cover proposal -> user intent -> broker observation -> journal -> post-close -> lesson lifecycle using synthetic MT5 records.
- TEST-007 [P1]** Browser E2E tests SHOULD cover authentication, analysis run/progress, cancellation, proposal display/user decision, MT5 offline/connected presentation, journal, performance, learning and settings.
- TEST-008 [P1]** CI SHOULD test supported Python versions 3.10-3.13, Ruff, clean install and browser E2E.
- TEST-009 [P1]** Static safety tests SHOULD assert no runtime broker order submission API is referenced/callable from the Forex code path.
- TEST-010 [P1]** Backup/restore tests MUST use temporary databases and verify restored records and integrity.
- TEST-011 [P1]** Tests MUST NOT be weakened merely to preserve an incorrect legacy behavior; requirement-driven behavior takes precedence.

30. Release States and Acceptance Gates

State	Meaning
ENGINEERING_COMPLETE	Required code paths implemented; automated unit/integration/E2E/CI gates pass.
READY_FOR_XM_DEMO	Engineering complete and documentation current; real broker acceptance still pending.
XM_DEMO_ACCEPTED	Mandatory Windows + XM demo checklist passed with evidence; still not proof of profitability.
STRATEGY_RESEARCH_VALIDATED	Separate statistical research standard, not implied by engineering or demo acceptance.
LIVE_USE_BY_USER	A user may choose to act manually; the software still provides no guarantee or autonomous execution.

REL-001 [P0] The project **MUST NOT** label itself XM/demo accepted until the mandatory real-terminal checklist has actually been executed.

REL-002 [P0] Passing automated tests **MUST NOT** be presented as proof of profitable trading performance.

REL-003 [P1] Release documentation **MUST** state the exact verified commit/test state or say verification is pending; it **MUST NOT** describe stale uncommitted history as current truth.

REL-004 [P1] The XM acceptance checklist **MUST** verify account/symbol/tick/positions/orders/deals, manual trade observation, modifications, partial/final close, costs, idempotency, post-close analytics and persistence across restart.

REL-005 [P0] The XM acceptance procedure **MUST** require all broker actions to be performed manually by the tester in MT5.

31. Fail-Closed Matrix and Error Semantics

Condition	Affected layer	Required result	Forbidden fallback
MT5 disconnected	MT5-backed analysis	DATA_INSUFFICIENT / connection required	No fake manual account
Account info unavailable	Live risk	NO actionable result	Do not substitute 100k/default balance
Quote unavailable/stale	Live risk	NO actionable result	No guessed spread/price
Positions unavailable	Portfolio risk	NO actionable result	Unknown != empty
Required conversion unavailable	Sizing	NO actionable result	No quote=account fallback
Broker spec incomplete	Live broker sizing	NO actionable result or clearly manual estimate	No silent generic live constraints
ATR unavailable	Spread/ATR rule	Explicit unavailable policy	Do not fabricate ATR
Calendar unavailable under mandatory blackout	Risk	NO actionable result or explicit degraded policy	Never silently assume no event
LLM analyst failure	Analysis	Run/stage fails explicitly	No fabricated report
Risk rejected	Final decision	NO_TRADE	LLM cannot override
Approved size <= 0	Executionability	NO_TRADE	Never replace with min/default lot
Historical conversion future/stale	Backtest	Data unavailable / skip action	No lookahead
Ambiguous SL/TP sequence	Backtest/metrics	AMBIGUOUS or conservative rule	No optimistic assumption
Post-close reflection provider failure	Learning	Persist FAILED/retryable	Metrics remain available

32. Non-Functional Requirements

- NFR-001 [P1]** The system SHOULD be understandable and maintainable by a small development team; business rules should live in domain/services rather than being duplicated across UI, routes and prompts.
- NFR-002 [P1]** Deterministic calculations SHOULD be reproducible: identical inputs/context should produce identical risk/sizing/metric outputs.
- NFR-003 [P1]** A normal live analysis SHOULD remain usable on an 8-16 GB RAM Windows machine; the system should avoid mandatory local heavyweight models or multiple worker processes.
- NFR-004 [P1]** Web requests that start long analyses SHOULD return promptly with a run ID while work continues asynchronously.
- NFR-005 [P1]** External provider timeouts/retries MUST be bounded so a run cannot hang indefinitely on routine HTTP failures.
- NFR-006 [P1]** Persistent records SHOULD be human-auditable through UI or exported representation without requiring direct SQLite inspection.
- NFR-007 [P1]** Errors should use stable categories/codes such as DATA_INSUFFICIENT, AUTH_REQUIRED, PROVIDER_ERROR, MT5_UNAVAILABLE, VALIDATION_ERROR and INTERNAL_ERROR.
- NFR-008 [P2]** Internationalization MAY cover reports/UI, but internal canonical enums/IDs MUST remain language-independent.
- NFR-009 [P1]** The system MUST avoid storing unnecessary personal/broker secrets in analytics, reports or learning records.

33. Practical Implementation Priorities

Priority group	What belongs here
Core Safety P0	Manual-only execution boundary; authoritative MT5/account/quote/portfolio context; one sizing authority; conversions; margin/free-margin; no-SL/unknown risk; daily loss; max positions; fail-closed semantics.
Core Workflow P1	Technical/Macro/News agents; debate; structured proposal; proposal lifecycle; browser workflow; read-only reconciliation; journal; post-close metrics; lessons.
Research Integrity P1	Point-in-time historical replay; next-eligible fills; walk-forward; cost assumptions; theoretical-vs-realized separation.
Operational P1	Security, auth, bounded runtime, cancellation, backups, CI, XM demo acceptance.
Enhancements P2	Monte Carlo, stop calibration, multi-hop FX conversion, extended exports, advanced lesson decay, optional remote/private deployment improvements.

What NOT to do first

Do not add more agents, auto-execution, complex cloud architecture, reinforcement learning, or strategy optimization before deterministic risk correctness, MT5 fail-closed behavior, historical causality and journal integrity are finished.

34. AI Implementation Rules and Conformance Process

- AI-001 [P0]** An AI coding model **MUST** treat this specification as target behavior, not infer requirements from incomplete current code.
- AI-002 [P0]** Before modifying code, the model **MUST** identify the requirement IDs being implemented and inspect the current code paths that own those responsibilities.
- AI-003 [P1]** The model **SHOULD** prefer the smallest coherent architectural change that creates one source of truth rather than adding another helper or duplicate calculation.
- AI-004 [P0]** The model **MUST** not add broker execution functionality even if it appears convenient for tests or demo automation.
- AI-005 [P0]** The model **MUST** not fabricate successful external validation, CI results, XM acceptance, data availability or performance claims.
- AI-006 [P1]** For each change, the model **MUST** add regression tests that demonstrate the requirement and the bug/failure mode being prevented.
- AI-007 [P1]** When a requirement conflicts with backward compatibility, financial correctness and safety take priority; the compatibility break must be documented.
- AI-008 [P1]** The model **SHOULD** report files changed, requirements satisfied, tests run, unresolved external dependencies and any requirement that remains only partially satisfied.
- AI-009 [P1]** The model **MUST** keep historical, manual-estimate and broker-verified states semantically distinct in API/UI/documentation.
- AI-010 [P1]** The model **MUST** stop and request human/external action where only real XM terminal behavior can validate the requirement.

Appendix A. Canonical Target Data Model

These schemas are target contracts, not a demand for exact class names. Existing Pydantic/dataclass models may satisfy them if the semantics are equivalent.

ForexRiskContext

- as_of_utc
- account profile
- broker constraints
- market quote/spread/ATR
- open positions
- pending exposures
- conversion observations
- daily realized P&L status/value
- source/provenance

ForexAccountProfile

- balance
- equity
- free_margin
- used_margin
- currency
- leverage
- policy ceilings

BrokerExecutionConstraints

- broker_symbol
- digits
- point
- pip_size
- contract_size
- volume_min
- volume_max
- volume_step

OpenPosition

- broker position id
- pair
- direction
- lots
- entry_price
- stop_loss optional
- risk_amount optional/unknown
- open_time

PendingExposure

- broker order id
- pair

- direction
- order_type
- remaining volume
- trigger price
- stop_loss optional
- reserved_risk optional/unknown
- setup_time

ForexConversionRate

- from_currency
- to_currency
- rate
- source
- observed_at
- path metadata

ForexTraderProposal

- proposal_id
- pair
- action
- order_type
- setup_type
- timeframe
- entry/zone
- SL
- TP1/TP2
- suggested risk
- approved size later
- confidence
- confluence
- invalidation
- reasoning
- decision time
- expiry
- run/snapshot/version metadata

ForexRiskDecision

- decision APPROVE/MODIFY/REJECT
- approved_action
- approved_lot_size
- risk percent
- pip value
- risk amount
- margin required
- checks passed
- violations
- modifications

- assumptions/context IDs

TradeRecord

- trade_id
- proposal_id optional
- pair
- direction
- status
- open/close times
- prices
- SL/TP
- lots
- commission
- swap
- fees
- gross/net P&L
- pips
- R
- exit reason
- metadata

ExecutionRecord

- deal/ticket IDs
- trade/proposal IDs
- pair
- order type
- volume
- price
- spread/slippage
- commission/swap/fee/profit where applicable
- timestamp

ForexLesson

- lesson_id
- source trade IDs
- pair/setup/timeframe/session/regime tags
- observation
- root cause
- actionable rule
- evidence class/count
- created/updated time
- recency/strength metadata

Appendix B. Detailed Workflow Scenarios

B1 Live MT5 analysis

User selects EURJPY/M15 and MT5 account. Server verifies terminal, account, EURJPYm spec, tick/spread, ATR, open positions, pending orders, daily P&L and JPY->USD conversion. Agents analyze one cutoff snapshot. Trader proposes. Deterministic risk/sizing approves/rejects. UI shows final result. No broker order is sent.

B2 Manual analysis

User supplies balance/equity/free margin/leverage/currency. System labels risk/sizing as manual/non-broker-verified and does not pretend current positions or broker constraints were checked unless explicitly supplied.

B3 User executes manually

After approval, user clicks EXECUTE intent then manually enters order in MT5. Observer later sees position/deal and matcher links it to the proposal. Journal becomes broker-backed.

B4 Unplanned manual trade

User opens a trade directly in MT5 without proposal. Observer detects it; journal creates MANUAL_UNPLANNED trade with no fake proposal.

B5 No-SL open position

Observer sees an existing position with no SL. Its bounded risk is unknown. Conservative portfolio policy blocks new directional risk rather than fabricating a 50-pip stop.

B6 Historical replay

At bar T close, system builds PIT snapshot and runs agents. Market proposal becomes eligible at T+1 first eligible price. Portfolio/account evolves through subsequent bars. Future news/lessons/conversions are never visible.

B7 Cancellation

User cancels an in-progress run. If cancellation wins before final guarded persistence, no final proposal/report is published. Repeated cancellation is safe.

B8 Provider failure after close

Trade metrics complete, reflection LLM fails. Post-close status becomes FAILED/retryable; deterministic metrics and journal records remain intact.

Appendix C. Explicitly Excluded or Deferred Features

Feature	Target decision
Automatic broker execution	Explicitly forbidden in target system.
Autonomous trade modification/close	Explicitly forbidden; observer only.
Guaranteed profitable signals	Impossible requirement and intentionally excluded.
HFT / millisecond execution	Outside human decision-support latency target.
Mandatory cloud microservices/Kubernetes	Unnecessary for single-user MT5 deployment.
Mandatory PostgreSQL	SQLite is sufficient for target deployment.
Self-modifying live strategy via RL	Too risky and unnecessary; learning is evidence-guided retrieval.
Tick-perfect historical fills without tick data	Not technically supportable from bar data; simulator must disclose precision.
Historical facts without publication timestamps	Must be unavailable rather than guessed.
LLM-calculated financial arithmetic as source of truth	Deterministic code is required.
Public Internet exposure by default	Local/private access is target deployment.
Large-scale multi-user RBAC SaaS	Not required for the personal/small-team target state.

Appendix D. Suggested Module Ownership Map

This appendix is implementation guidance only. It maps target responsibilities onto the current repository structure without making the repository itself authoritative.

Responsibility	Likely implementation area
Forex domain / symbols / timeframes	tradingagents/forex/domain.py, symbols.py
Indicators / structure / sessions	tradingagents/forex/indicators.py, market_structure.py, sessions.py
Conversion / pip arithmetic	tradingagents/forex/conversion.py, pips.py
Risk context / policy / sizing	tradingagents/risk/context.py, engine.py, sizing.py
Agent graph	tradingagents/graph/forex_graph.py and Forex agents
MT5 read-only adapter	tradingagents/mt5/connection.py, observer.py, service.py, models.py
Contracts / run records	tradingagents/research/contracts.py
Journal / lifecycle / matching	tradingagents/database/, tradingagents/journal/
Post-close metrics	tradingagents/metrics/
Learning	tradingagents/learning/
Performance / calibration / Monte Carlo	tradingagents/analytics/
Historical replay	tradingagents/backtest/
Web application	web/forex_routes.py, web/forex_runtime.py, web/static/
CLI	cli/main.py and supporting CLI modules
Tests	tests/ with focused unit, integration and browser E2E coverage

Final Target-State Acceptance Checklist

- CHECK-01** No runtime MT5 broker-order submission path exists.
- CHECK-02** Live MT5 analysis fails closed on unknown account/quote/position/conversion/broker state.
- CHECK-03** Exactly one authoritative sizing result is used by the normal graph.
- CHECK-04** Cross-pair and nontrivial account-currency sizing is regression-tested.
- CHECK-05** Open positions, pending exposure, daily loss and portfolio ceilings affect new risk.
- CHECK-06** No-SL existing positions are handled conservatively.
- CHECK-07** Proposal lifecycle is separate from broker execution.
- CHECK-08** MT5 observation and journal reconciliation are idempotent.
- CHECK-09** Post-close metrics, reflection and lesson retrieval work with explicit retry/unavailable states.
- CHECK-10** Historical analysis has point-in-time integrity and next-eligible-price execution.
- CHECK-11** Walk-forward/OOS logic does not leak future data.
- CHECK-12** UI, CLI and API label broker-verified vs manual-estimate vs unavailable states truthfully.
- CHECK-13** Secrets are not persisted or exposed.
- CHECK-14** SQLite backup/restore is tested.
- CHECK-15** Automated unit/integration/E2E/CI gates pass.
- CHECK-16** Windows + XM Demo manual acceptance is executed before claiming broker acceptance.
- CHECK-17** No engineering gate is described as proof of profitability.

END OF NORMATIVE SPECIFICATION

This target-state document intentionally defines a feasible, human-in-the-loop Forex research and risk platform rather than an autonomous trading system. Future implementation work should be judged against these requirements, not against the accidental behavior of an incomplete repository.